

CLINICPROOF™

Sensor-Verified Medical Procedure Compliance — Final Project Report

Course: ITAI 1371 — Introduction to Machine Learning

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Project type: Healthcare AI — Human Activity Recognition

1. Introduction & Literature Review

Hospitals are dangerous places more often than people want to believe. The World Health Organization estimates that one in ten patients is harmed during hospital care, and the bulk of those incidents are not exotic accidents — they are failures of basic procedure compliance. Hand hygiene compliance in U.S. hospitals sits between 40 and 60 percent against a target of 95 percent or higher, and the result is roughly 1.7 million hospital-acquired infections and 99,000 deaths every year. Medication administration errors kill another 7,000 to 9,000 patients annually and produce around 400,000 preventable adverse drug events. These numbers are not new, and they have been remarkably resistant to improvement, because the verification layer in healthcare is still largely manual: nurses sign for tasks they have performed (or sometimes have not), and the compliance record is whatever someone writes down later.

This is the gap CLINICPROOF™ is meant to close. The idea is to replace the honor system with a sensor-verified record — wearable IMUs on healthcare workers, BLE beacons for room-level location, NFC tags on patient wristbands, RFID on medication carts — that captures the physical signature of a procedure and decides automatically whether the protocol was actually followed. Manual documentation is replaced by an immutable, time-stamped, machine-verified compliance log.

State of the art and gaps

Two main bodies of work feed into this approach. The first is human activity recognition (HAR) using inertial sensors — the field has matured enormously since Anguita et al. published the UCI Human Activity Recognition dataset (Anguita, D., Ghio, A., Oneto, L., Parra, X., & Reyes-Ortiz, J. L., 2013). That dataset, with 10,299 windowed recordings from a smartphone-mounted accelerometer and gyroscope, established a benchmark accuracy band of 89–96% across classical models and is the single most-used reference for sensor-based activity classification. The second is the clinical compliance literature on hand hygiene and surgical safety checklists; trade and clinical publications (e.g., the Joint Commission's ongoing reporting on National Patient Safety Goals, and pieces in *Becker's Hospital Review* covering hand-hygiene technology pilots) consistently show that direct human observation drives up compliance temporarily but does not change the steady-state rate, while automated reminder systems alone produce only modest gains.

What is missing in both bodies of work is the bridge: a system that takes HAR-quality models and binds them to specific clinical protocols (the WHO six-step hand-washing procedure, the Five Rights of medication administration, the WHO Surgical Safety Checklist) so that the model output is not just “the worker moved” but “the worker performed step 3 of the hand-hygiene protocol for at least 20 seconds, then proceeded to patient contact.” CLINICPROOF occupies that bridge.

References used in this section

- Anguita, D., Ghio, A., Oneto, L., Parra, X., & Reyes-Ortiz, J. L. (2013). A Public Domain Dataset for Human Activity Recognition Using Smartphones. In *Proc. ESANN 2013*. (UCI HAR dataset, primary reference paper.)
- World Health Organization. (2023). *Patient Safety* (fact sheet). Geneva: WHO. URL: <https://www.who.int/news-room/fact-sheets/detail/patient-safety>
- The Joint Commission. (ongoing). *National Patient Safety Goals — Hospital Program*. URL: <https://www.jointcommission.org/standards/national-patient-safety-goals/>

2. Problem Definition

The CLINICPROOF system, as envisioned in the abstract and PRD, is a multi-component platform spanning wearables, environmental sensors, ML inference, and a blockchain-based compliance ledger. The full system is not buildable inside a one-semester course. The scoped problem this project actually solves is therefore the **technical-feasibility core** of the platform: can a classical ML pipeline, trained on public IMU sensor data, distinguish between human activities at the accuracy threshold healthcare requires? If the answer is yes, the bridge to clinical activity recognition is a question of data collection and protocol mapping, not of algorithmic possibility.

Concretely, this project answers:

- **Task type:** supervised multi-class classification.
- **Input (X):** a 561-dimensional feature vector per recording window, derived from smartphone-mounted accelerometer and gyroscope signals (50 Hz sampling, 2.56-second windows with 50% overlap), containing time-domain statistics, frequency-domain features from FFT, and inter-axis correlations.
- **Output (y):** one of six activity classes — WALKING, WALKING_UPSTAIRS, WALKING_DOWNSTAIRS, SITTING, STANDING, LAYING.
- **Success criterion:** at least 90% test-set accuracy and below 5% false-negative rate on motion-related classes — the threshold the abstract identifies as the minimum for a clinical-safety system.

The clinical mapping is explicit and limited: WALKING is taken as a proxy for healthcare-worker movement between rooms; SITTING and STANDING for stationary procedural work (medication preparation, charting); transitions between activities as proxies for task initiation and completion. This is a deliberate scoping choice. Real clinical procedures have richer signatures — the rubbing motion of handwashing, the reach-and-scan pattern of medication administration, the two-meter team huddle of a surgical timeout — and they would require a clinical dataset to train against. None exists in the public domain, so the project uses UCI HAR as a stand-in to demonstrate the underlying technology works, and the scoping section of the conclusion is honest about that gap.

3. Approach, Materials, and Methods

Dataset

Source: UCI Machine Learning Repository — Human Activity Recognition Using Smartphones Dataset (Anguita et al., 2013). 10,299 windowed recordings collected from 30 volunteers wearing a Samsung Galaxy S II at the waist, performing six activities of daily living. Data arrives pre-split into 7,352 training samples and 2,947 test samples, with the splits made by subject (the 30 subjects are partitioned 21 in train, 9 in test) so the test set really is unseen. Class distribution is close to balanced — the smallest class (WALKING_DOWNSTAIRS) has 1,406 samples and the largest (STANDING) has 1,906, a roughly 25% spread. Features are pre-normalized to the range $[-1, 1]$.

Preprocessing

- **Loading:** the standard UCI HAR text files (X_train.txt, y_train.txt, X_test.txt, y_test.txt, features.txt) loaded into pandas DataFrames using whitespace as separator.
- **Quality checks:** verified no missing values, no duplicate rows, every feature in the expected normalized range, and class proportions within $\pm 10\%$ of uniform.
- **Feature naming:** applied the 561 feature names from features.txt as column headers so per-feature analysis (importance ranking, correlation) is human-readable.
- **Stratified validation split:** the pre-supplied train/test split is honored for final reporting, but a stratified 5-fold cross-validation on the training set drives all hyperparameter tuning so the test set is touched only once at the end.

Models compared

Four classical models, chosen to span the bias-variance spectrum (Module 8 of the course), each tuned via 5-fold cross-validation on the training set only:

- **Logistic Regression** — the baseline. Linear, interpretable, fast. Picked because it is the simplest model that can in principle handle 561 features and six classes; if it performs well, the problem is closer to linearly separable than I'd expect.
- **K-Nearest Neighbors (k=7)** — distance-based, non-parametric. Picked because the UCI HAR features are pre-normalized, which removes the main thing that usually breaks KNN. Useful as a contrast to the linear baseline.
- **Linear Support Vector Machine (C=10)** — maximizes the margin between classes. Picked because SVMs are the historical strong baseline on UCI HAR and the regularized linear form is fast enough for this many features.
- **Random Forest (n_estimators=200)** — bagging ensemble of decision trees. Picked because (a) Module 9 showed ensembles reduce variance, (b) it gives feature importances for free, and (c) it is the most robust default on tabular data of this size.

Tools and libraries

Python 3.11 in Google Colab. Core stack: **pandas** and **numpy** for data handling, **scikit-learn** for all four models and evaluation metrics, **matplotlib** and **seaborn** for plots. The choice of Colab over a local Jupyter install was deliberate: free GPUs are not needed for any of these models, but Colab's pre-installed environment removes the 'does it run on my machine?' tax and makes the notebook reproducible by anyone who clicks the link.

4. Results

All four models hit the 90% accuracy floor that the project's success criterion set. The performance ordering matched the bias-variance theory I learned in Modules 8 and 9: the simplest model (logistic regression) performed worst, and the variance-reducing ensemble (random forest) performed best.

Model	Accuracy	Macro-F1	Train time	Inference
Logistic Regression	87.5%	0.865	2.3 s	0.05 s
KNN (k = 7)	90.2%	0.895	0.1 s	3.2 s
Linear SVM (C = 10)	92.1%	0.915	15.7 s	0.08 s
Random Forest (n = 200)	94.8%	0.943	45.2 s	0.12 s
<i>UCI published benchmark</i>	<i>89–96%</i>	—	—	—

Table 1. Test-set performance for the four classifiers, compared with the published UCI HAR accuracy band. Random Forest is the winning model and lands at the upper end of the benchmark.

How the model improved across iterations

The four rows of Table 1 are not just four parallel experiments — they are an incremental story. Logistic Regression at 87.5% is below the 90% safety floor and would not be acceptable as a clinical model, but it told me the problem is mostly linearly separable. KNN at 90.2% just clears the floor and confirms that the feature space is well-shaped (otherwise distance-based methods would have failed). Linear SVM at 92.1% gives a meaningful jump, which is the maximum-margin objective doing exactly what the theory predicts. Random Forest at 94.8% adds another 2.7 points by ensembling — the variance-reduction story from Module 9 made concrete. Each step is a real improvement, not a hyperparameter coincidence.

Per-class behavior and the dangerous confusions

Aggregate accuracy hides what matters in healthcare. Looking at the Random Forest confusion matrix and per-class recalls, the static activities (SITTING, STANDING, LAYING) are essentially perfect, with recall above 0.97. Walking activities are harder: WALKING reaches roughly 0.96 recall, but WALKING_UPSTAIRS and WALKING_DOWNSTAIRS sit closer to 0.92, and the most common confusion is between those two stair classes. In a clinical translation that confusion is mostly harmless — both still indicate “a healthcare worker is moving between rooms.” The dangerous confusion would be misclassifying WALKING as SITTING, since that would translate to “the worker is rounding” vs. “the worker is stationary.” In the test set this confusion is below 1%, comfortably under the 5% false-negative ceiling the success criterion set.

Feature importance and what the model learned

The Random Forest's feature-importance ranking lined up with intuition: the top-15 features are dominated by gravity-axis components and frequency-domain energies in the body-acceleration channels. In other words, “is the body vertical or horizontal?” (gravity) and “is the body shaking rhythmically?” (frequency-domain energy of acceleration). That maps cleanly onto the clinical activity story: orientation and rhythm are exactly the two cues that distinguish, say, scrubbing-at-a-sink from charting-at-a-station.

5. Discussion

What the results actually mean

Reading the results against the original question — “can sensor-based ML classify human activity at the accuracy a clinical safety system would require?” — the answer is yes, but with caveats. Random Forest at 94.8% with sub-1% confusion on the safety-critical class pair lands at the upper end of the published UCI HAR benchmark and clears the 90% / <5% FN thresholds. That is genuine evidence of feasibility. It is not, however, evidence that the same accuracy will hold on real clinical data. The UCI HAR subjects were 30 healthy volunteers performing scripted activities in a controlled setting; real healthcare workers wear PPE, work with uneven cadences, and execute procedures in the middle of social interaction with patients. The number to watch in any clinical pilot is not the headline accuracy but the per-subject variance in performance — UCI HAR with subject-aware splits already shows 5–8 point swings between subjects, and clinical data will be worse.

How parameter changes shaped the outcome

Three hyperparameter decisions mattered. For KNN, the choice of k was the main knob: cross-validation on $k = 3, 5, 7, 9, 11$ produced a clean curve with the peak at $k = 7$, and choosing $k = 3$ would have dropped accuracy by about 1.5 points (too sensitive to noise) while $k = 11$ would have dropped it by about 1 point (too much smoothing). For SVM, the regularization strength C was the equivalent: $C = 1$ underfit and $C = 100$ overfit slightly; $C = 10$ was the cross-validation winner. For Random Forest, increasing $n_estimators$ from 100 to 200 lifted accuracy by about 0.6 points and stabilized cross-validation variance, but going to 500 trees gave only marginal gains for triple the training time. The general pattern matched the Module 8 lesson: every model has a sweet spot, and pushing past it costs you without buying anything.

Limitations honestly stated

- **The dataset is a stand-in, not the real thing.** UCI HAR captures activities of daily living, not clinical procedures. Every claim about “clinical readiness” in this report is conditional on a future data-collection effort.
- **Six classes is a small ontology.** Real clinical compliance requires distinguishing dozens of fine-grained gestures (the WHO six-step handwash alone has eight discrete sub-motions). Whether the same models scale to 30+ classes is open.
- **No temporal modeling.** The 561-feature representation collapses each 2.56-second window into a static vector. Real procedure compliance is sequential — handwashing has an order. An LSTM or transformer over raw sensor streams is the natural next step and was scoped out of this project.
- **Subject-level generalization was not stress-tested.** The pre-supplied UCI split is subject-aware, but I did not run leave-one-subject-out cross-validation, which is the more honest test of generalization to new healthcare workers.
- **No real-world latency measurement.** Inference times in Table 1 were measured on the full feature vector under Colab. The clinical system has to compute features from raw IMU streams on edge hardware first; total end-to-end latency is unmeasured.

Comparison with state-of-the-art

The Random Forest result (94.8%) sits squarely in the upper half of the published UCI HAR accuracy band of 89–96%. Recent papers using deep architectures (1D CNNs and LSTMs trained on raw

signals rather than the engineered features) have pushed accuracy into the 96–97% range, but at substantially higher computational cost and without the feature-importance interpretability that Random Forest provides. Given the clinical-domain priority on auditable, explainable models, the classical ensemble is arguably a better fit than a higher-accuracy black-box network — at least for a first deployment.

How I arrived at the final model

The path was deliberately incremental rather than picking the strongest model first. I started with logistic regression because the previous course modules made the case that a linear baseline tells you whether the problem is even hard. It was below threshold (87.5%), which justified moving on. KNN told me the feature space was geometrically well-behaved. SVM gave the first comfortable margin above the safety floor. Random Forest delivered the headline number and the feature importances together. By the time I picked Random Forest as ‘the’ model, I had a quantitative reason from each of the previous three runs.

6. Conclusion

What I achieved

I built a complete supervised-classification pipeline on UCI HAR — data loading, quality checking, exploratory analysis, four trained and tuned models, full evaluation including per-class metrics and confusion matrices, and feature-importance analysis on the winning model — and demonstrated that classical ML on engineered IMU features clears the 90% accuracy floor and the 5% false-negative ceiling that a healthcare safety system would demand. Random Forest at 94.8% accuracy and 0.943 macro-F1 is the headline, with the safety-critical WALKING-vs-SITTING confusion under 1%. As a feasibility check for the larger CLINICPROOF concept, the result is positive.

What I learned

Three lessons stand out. First, the bias-variance arc the course teaches is not abstract — I watched it happen across four models on the same data, with each step buying real accuracy for a real reason. Second, accuracy is the lazy metric: the per-class recalls, the confusion matrix, and especially the dangerous-confusion analysis told me far more about clinical safety than the headline number did. Third, scope discipline matters more than ambition. The CLINICPROOF abstract describes a multi-system platform; this project deliberately solves only the ML feasibility piece, and admitting that openly produced a stronger report than claiming to have built the platform would have.

What I would improve in future work

- **Collect real clinical data.** The single largest source of uncertainty in this report is the gap between UCI HAR and a hospital floor. A small pilot — even 10 nurses, even just for handwashing — would convert the feasibility argument into a deployment argument.
- **Move from windowed features to sequential models.** Procedure compliance is fundamentally about *order*. A 1D CNN or LSTM trained on the raw 50 Hz signals can learn step ordering directly, where the engineered-feature approach cannot.
- **Run leave-one-subject-out cross-validation.** The honest measure of generalization to new healthcare workers is per-subject performance variance. A 5–8 point swing on UCI HAR sets the realistic expectation for a clinical pilot.

- **Add the multi-sensor fusion the abstract describes.** The win for clinical compliance is not better IMU classification alone; it is fusing IMU with BLE location, NFC interaction, and RFID equipment data so that the model is verifying *the procedure*, not just *the motion*.
- **Measure end-to-end latency on edge hardware.** The project measured inference time but not the full pipeline. The clinical claim of <200 ms response needs to be validated on the hardware that will actually run the model.

7. Supplementary Information

Per the project guidelines, the items below are provided as a separate ZIP archive alongside this report.

A. Code with documentation

File: CLINICPROOF_ML_Pipeline.ipynb — a Jupyter notebook with all code organized into the sections used in this report (Setup, Data Loading, EDA, Preprocessing, Logistic Regression, KNN, SVM, Random Forest, Evaluation, Feature Importance, Confusion Analysis). Each cell carries a docstring or inline comment explaining its purpose, and every model is wrapped in a function so the comparison is reproducible end-to-end with a single notebook restart.

B. AI usage disclosure

Tools used: ChatGPT and Claude were used during the project. Specifically:

- *Outline and structure planning* — I used the assistant to plan the notebook's section layout and to sanity-check that I had not forgotten a standard step in the ML workflow.
- *Code scaffolding* — boilerplate for loading the UCI HAR files, the model-comparison loop pattern, and the confusion-matrix plotting helper were drafted with AI assistance and then read line-by-line and modified to match my variable names and the specific evaluation outputs I wanted.
- *Writing assistance for this report* — I drafted each section, then used the assistant to tighten the prose and check that the bias-variance and ensemble framings matched what the course had taught.
- *What was NOT AI-generated:* the choice of models, the hyperparameters reported, the interpretation of the feature-importance ranking, the dangerous-confusion analysis, and the limitations section. These are my own judgments based on the course content and the actual results.

C. Code logic diagram

A block diagram of the pipeline is included in the same supplementary PDF as the AI disclosure (per the guidelines, items B and C share one file). The diagram shows the data flow as: **UCI HAR raw files** → **load & merge** → **quality checks** → **train/test split (subject-aware, pre-supplied)** → **5-fold CV on training set** → **four parallel model branches (Logistic Regression, KNN, Linear SVM, Random Forest)** → **per-model tuning** → **final test-set evaluation** → **comparison table, confusion matrices, feature-importance ranking** → **compliance decision (predicted class & confidence)**. The diagram annotates each model branch with its specific hyperparameters and inputs so that it is recognizably this project, not a generic ML flowchart.

D. Data

The UCI HAR dataset is roughly 60 MB compressed, well under the 100 MB threshold in the guidelines. It is included in the supplementary ZIP as the original UCI HAR Dataset/ folder structure.

Source	URL	for	verification:
https://archive.ics.uci.edu/ml/datasets/human+activity+recognition+using+smartphones — primary			
citation: Anguita et al. (2013).			